

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit III | Session 1 | SLO 2: Thread Lifecycle and States (Code: 3012)

Q: 1. Fill in the blank:

A thread in Java is created in the _____ state before start() is invoked.

Ans: Answer: NEW

Q: 2. True or False:

Calling run() directly creates a new operating system thread.

Ans: Answer: False (Calling run() executes the method in the caller's thread; start() creates a new thread).

Q: 3. Multiple Choice:

Which method must be called to begin thread execution?

A) run() B) start() C) init() D) execute()

Ans: Answer: B) start()

Q: 4. Fill in the blank:

When a thread is waiting to acquire a monitor lock, it is in the _____ state.

Ans: Answer: BLOCKED

Q: 5. Programming – Output:

```
Thread t = new Thread(() -> System.out.println("Thread running"));  
t.start();
```

Ans: Output: Thread running

Q: 6. True or False:

A terminated thread can be restarted by calling start() again.

Ans: Answer: False (Throws IllegalStateException).

Q: 7. Multiple Choice:

When the run() method finishes executing, the thread transitions to which state?

A) WAITING B) RUNNABLE C) TERMINATED D) BLOCKED

Ans: Answer: C) TERMINATED

Q: 8. Fill in the blank:

The method Thread._____(ms) pauses execution for specified milliseconds.

Ans: Answer: sleep

Q: 9. Identify Error:

```
Thread t = new Thread(); t.start(); t.start();
```

Ans: Answer: Throws IllegalStateException at runtime.

Q: 10. Fill in the blank:

The exact scheduling order of multiple runnable threads is _____.

Ans: Answer: non-deterministic (or determined by OS thread scheduler)